

# Mittag-Leffler Stability Analysis for a fractional order model of co-contaminated soil and water pollution

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**Abstract:** Environmental pollution is a serious and long-lasting problem, and understanding how it spreads and how it can be controlled requires robust mathematical models. In this work, we study a five-compartment fractional-order model that describes how pollution moves between different environmental components, namely effluent, soil, water, and farmland. To capture the memory and hereditary effects that naturally occur in environmental processes, we use the Atangana–Baleanu–Caputo (ABC) fractional derivative, which provides a more realistic description than classical integer-order models. We first analyze the qualitative behavior of the model. Using linearization, Lyapunov function, and tools from fractional calculus, we establish the local and global asymptotic stability of two important equilibrium points: the pollution-free equilibrium (E) and the endemic (polluted) equilibrium (E\*). In particular, we apply the concept of Mittag-Leffler stability, which is well-suited to fractional-order systems. A central result of the analysis is the derivation of the basic reproduction number

**Keywords:** Co-Contamination, ABC mathematical model, fixed point, Adaptive control, Adams-Bhashforth

## 1. Introduction

Environmental pollution affecting soil and water systems has emerged as a serious global challenge due to industrial discharge, agricultural runoff, and uncontrolled waste disposal. Since soil and water environments are strongly coupled, contaminants introduced into one medium can easily migrate to the other, leading to persistent ecological damage and long-term health risks. Mathematical modeling is an essential tool for understanding such complex pollution dynamics and for designing effective remediation strategies. However, classical integer-order models often fail to capture the memory effects, nonlocal interactions, and anomalous transport mechanisms observed in real environmental processes. Fractional calculus provides a powerful and flexible framework for modeling systems with memory and hereditary properties. Fractional-order differential equations have been successfully applied in various scientific fields, including heat transfer, diffusion, epidemiology, groundwater flow, and ecological systems (Diethelm, 2010; Baleanu et al., 2019). Unlike integer-order models, fractional models are capable of representing long-range temporal dependence and anomalous diffusion. In recent years, special attention has been given to fractional derivatives with nonlocal and non-singular kernels. Among them, the Atangana–Baleanu fractional derivative in Caputo sense has attracted considerable interest due to its Mittag-Leffler kernel, which avoids singularities and provides better physical interpretation (Atangana & Baleanu, 2017). This operator has been widely applied to heat transfer models and groundwater flow systems (Atangana & Algahtani, 2015; Sun et al., 2019). Efficient numerical methods for solving fractional-order systems with the Atangana–Baleanu–Caputo derivative have been actively developed, showing good stability and convergence properties (Yadav & Pandey, 2019; Singh et al., 2022; Yadav & Pandey, 2020). Stability analysis is fundamental in fractional-order dynamical systems, where stability is governed by Mittag-Leffler functions rather than exponential decay. Rigorous stability criteria and Lyapunov-based methods have been developed for such systems (Li & Zhang, 2019; Vargas-De-León, 2012). Motivated by these developments, this study proposes a fractal fractional-order model for co-contaminated soil and water pollution using the Atangana–Baleanu–Caputo derivative. The local and

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global stability of the system equilibria is analyzed using Mittag-Leffler stability theory, supported by numerical simulations.

This study contributes to the field by presenting a detailed and rigorous analysis of a five-compartment fractional-order model for pollution dynamics and remediation, specifically utilizing the Atangana-Baleanu-Caputo derivative. We extend previous work by providing a thorough local and global stability analysis of both the pollution-free (trivial) and persistent (endemic) equilibrium points. Our analysis employs advanced techniques such as Mittag-Leffler stability and fractional Lyapunov methods to characterize the system's long-term behavior precisely. Furthermore, we develop and implement an advanced numerical scheme tailored for the efficient and accurate simulation of the ABC fractional system, enabling practical exploration of model behavior and validation of the theoretical findings. This comprehensive approach, combining rigorous theoretical analysis with robust computational tools, aims to provide deeper insights into the conditions necessary for successful pollution remediation and the long-term behavior of complex, memory-dependent environmental systems.

## 2. Mathematical Preliminaries

### 2.1 Fractional Calculus Fundamentals

Definition 2.1: (Atangana-Baleanu-Caputo Fractional Derivative). [?] For  $0 < \alpha < 1$ , the Atangana-Baleanu-Caputo fractional derivative of order  $\alpha$  is defined as:

$${}^{ABC}D_t^\alpha f(t) = \frac{B(\alpha)}{1-\alpha} \int_0^t E_\alpha \left( -\alpha \frac{(t-\tau)^\alpha}{1-\alpha} \right) f'(\tau) d\tau,$$

where  $B(\alpha)$  is a normalization function with  $B(0) = B(1) = 1$ , and  $E_\alpha(\cdot)$  is the Mittag-Leffler function.

Definition 2.2: (Mittag-Leffler Function).[?] The one-parameter Mittag-Leffler function is defined as:  $E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + 1)}$ ,  $\alpha > 0, z \in \mathbb{C}$ .

The two-parameter Mittag-Leffler function is:

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta)}, \quad \alpha, \beta > 0, z \in \mathbb{C}.$$

### 2.2 Mittag-Leffler Stability Theory

Definition 2.3 (Mittag-Leffler Stability) [?]. The solution  $x(t)$  of a fractional-order system is said to be Mittag-Leffler stable if:  $\|x(t)\| \leq [m(x_0) E_\alpha(-\lambda t^\alpha)]^b$  where  $x_0$  is the initial condition,  $m(0) = 0$ ,  $m(x) \geq 0$ ,  $\lambda > 0$ , and  $b > 0$  are constants.

Remark 2.1. When  $\alpha = 1$ , Mittag-Leffler stability reduces to exponential stability since  $E_1(-\lambda t) = e^{-\lambda t}$ . For  $0 < \alpha < 1$ , the decay is typically slower than exponential, reflecting the memory effects in fractional systems.

Lemma 2.1 (Fractional Comparison Principle). [?] Let  $u(t) \in C^1([0, T], \mathbb{R}_+)$  satisfy:  ${}^{ABC}D_t^\alpha u(t) \leq -\lambda u(t)$ ,  $u(0) = u_0$  Then:  $u(t) \leq u_0 E_\alpha(-\lambda t^\alpha)$

## 3. Model Formulation

**Consider the fractional-order system with Atangana-Baleanu-Caputo derivatives:**

$$tABCC = aN - LCS_p - BCW - \Phi_C C$$

$$tABCS_p = LCS_p - \delta_1 S_p W - \delta_2 S_p F - \Phi_2 S_p - \gamma_1 S_p$$

$$tABCW = \delta_1 S_p W - \rho WF - \Phi_3 W$$

$$tABCF = \delta_2 S_p F + \rho WF - \gamma_2 F - \Phi_4 F$$

$$tABCS_R = \gamma_1 S_p + \gamma_2 F - \Phi_5 S_R$$

(1.1)

With the initial conditions

$$C^0 = C(0) \geq 0, S_p^0 = S_p(0) \geq 0, W^0 = W(0) \geq 0, F^0 = F(0) \geq 0, S_R = S_R(0) \geq 0.$$

$a$  – Effluent Concentration

$N$  – Total Area

$L$  – Rate of contaminated sediments in soil

$\beta$  – Rate of contaminated settlements in water  
 $\delta_1$  – Washout rate of effluents from soil into water  
 $\delta_2$  – Polluted settlements in farmland  
 $\rho$  – Polluted water affecting farmland  
 $\gamma_1$  – Remedy in polluted soil  
 $\gamma_2$  – remedial measure in farmland  
 $\phi_1, \phi_2, \phi_3, \phi_4, \phi_5$  – Mortality rate of all the compartments  
 $C$  – Effluent Concentration  
 $S_p$  – Polluted Soil  
 $W$  – Water Contamination  
 $F$  – Farmland pollution  
 $S_R$  – Remedied Soil

Definition 3.4. [?] The system is said to be locally asymptotically stable at an equilibrium point  $E^*$  if all eigenvalues  $\lambda$  of the Jacobian matrix satisfy  $|\arg(\lambda)| > \frac{\alpha\pi}{2}$ .

Definition 3.5.[?] The system is globally asymptotically stable at an equilibrium point  $E^*$  if it is stable and all trajectories approach  $E^*$  as  $t \rightarrow \infty$ .

#### 4. Equilibrium Point and Stability analysis:

The equilibrium points for the fractional-order pollution remediation model are found by setting the right-hand sides of the model equations (1.1) to zero.

Trivial Equilibrium Point ( $E_0$ ):

This equilibrium represents a state where pollution is absent in the soil, water, and farmland compartments.

$$E_0 = \left( \frac{aN}{\phi_C}, 0, 0, 0, 0 \right).$$

The components are:

$$C_0 = \frac{aN}{\phi_C}$$

): Effluent concentration reaches a baseline level.

$S_{p0} = 0$ : No polluted soil.

$W_0 = 0$ : No water contamination.

$F_0 = 0$ : No farmland pollution.

$S_{R0} = 0$ : No remediated soil.

Endemic Equilibrium Point ( $E^*$ ):

This equilibrium represents a state where pollution persists in all environmental compartments.

$$E^* = (C^*, S_p^*, W^*, F^*, S_R^*).$$

where  $C^* > 0$ ,  $S_p^* > 0$ ,  $W^* > 0$ ,  $F^* > 0$ , and  $S_R^* > 0$ . The specific values are determined by solving the system of algebraic equations and we obtained,

$$\begin{aligned}
 C^* &= \frac{(aN - LC_S - BC_W)}{\phi_C} \\
 S_p^* &= \frac{LC_S}{(\delta_1 W + \delta_2 F + \phi_2 + \gamma_1)} \\
 W^* &= \frac{\delta_1 S_p}{(\rho F + \phi_3)} \\
 F^* &= \frac{(\delta_2 S_p + \rho W)}{(\gamma_2 + \phi_4)} \\
 S_R^* &= \frac{(\gamma_1 S_p + \gamma_2 F)}{\phi_5}.
 \end{aligned}$$

#### 4.2 Stability analysis

In this section, we analyze the stability properties of the fractal-fractional order model for co-contaminated soil and water pollution. The system is governed by the Atangana-Baleanu-Caputo (ABC) fractional derivative, which enables the incorporation of memory effects and non-local dynamics into the pollution transmission process.

### 4.2.1 Local Stability Analysis

Jacobian Matrix Computation

Let  $X = (C, S_p, W, F, S_R)^T$ . The Jacobian matrix  $J(E)$  evaluated at equilibrium point  $E = (C^*, S_p^*, W^*, F^*, S_R^*)$  is:

$$J(E) = \begin{pmatrix} -\Phi_C - LS_p^* - BW^* - LC^* - BC^* & 0 & 0 & LS_p^* & LC^* - \delta_1 W^* - \delta_2 F^* - \Phi_2 - \gamma_1 & -\delta_1 S_p^* - \delta_2 S_p^* & 0 & 0 & \delta_1 W^* & \delta_1 S_p^* - \rho F^* - \Phi_5 \end{pmatrix}$$

### Local Stability of Trivial Equilibrium

Proposition 4.1. The trivial equilibrium point  $E_0 = \left(\frac{aN}{\Phi_C}, 0, 0, 0, 0\right)$  has Jacobian matrix:

$$J(E_0) = \begin{pmatrix} -\Phi_C - L\frac{aN}{\Phi_C} - B\frac{aN}{\Phi_C} & 0 & 0 & 0 & L\frac{aN}{\Phi_C} - \Phi_2 - \gamma_1 & 0 & 0 & 0 & 0 & -\Phi_3 & 0 & 0 & 0 & 0 & -\gamma_2 - \Phi_4 & 0 & 0 & \gamma_1 & 0 & \gamma_2 & -\Phi_5 \end{pmatrix}$$

Theorem 4.1. The trivial equilibrium  $E_0$  is locally asymptotically stable if:  $L\frac{aN}{\Phi_C} < \Phi_2 + \gamma_1$

Proof. The characteristic equation is:

$$\det(J(E_0) - \lambda I) = (-\Phi_C - \lambda)(-\Phi_3 - \lambda)(-\Phi_5 - \lambda)(-\gamma_2 - \Phi_4 - \lambda) \left( L\frac{aN}{\Phi_C} - \Phi_2 - \gamma_1 - \lambda \right) = 0$$

The eigenvalues are:

- $\lambda_1 = -\Phi_C < 0$
- $\lambda_2 = -\Phi_3 < 0$
- $\lambda_3 = -\Phi_5 < 0$
- $\lambda_4 = -(\gamma_2 + \Phi_4) < 0$
- $\lambda_5 = L\frac{aN}{\Phi_C} - \Phi_2 - \gamma_1$

For stability, we require  $\lambda_5 < 0$ , which gives the condition:

$$L\frac{aN}{\Phi_C} < \Phi_2 + \gamma_1$$

Since  $0 < \alpha < 1$ , we have  $\frac{\alpha\pi}{2} < \frac{\pi}{2}$ , so  $\cos\left(\frac{\alpha\pi}{2}\right) > 0$ . For real eigenvalues, the stability condition  $|\arg(\lambda)| > \frac{\alpha\pi}{2}$  is equivalent to  $\lambda < 0$  when  $\lambda$  is real and negative.

Local Stability of Endemic Equilibrium

Theorem 4.2 (Local Stability of Endemic Equilibrium). The endemic equilibrium point  $E^*$  is locally asymptotically stable if all eigenvalues  $\lambda$  of the Jacobian matrix  $J(E^*)$  satisfy  $|\arg(\lambda)| > \frac{\alpha\pi}{2}$  where  $0 < \alpha < 1$  is the fractional order.

Proof. The characteristic equation is  $\det(J(E^*) - \lambda I) = 0$ . Due to the triangular structure of the last row, one eigenvalue is  $\lambda_5 = -\Phi_5 < 0$ , which is always stable.

For the remaining  $4 \times 4$  subsystem, let:

$$J_4 = \begin{pmatrix} -\Phi_C - LS_p^* - BW^* - LC^* - BC^* & 0 & LS_p^* & 0 & -\delta_1 S_p^* - \delta_2 S_p^* & 0 & \delta_1 W^* & 0 & -\rho W^* & 0 & \delta_2 F^* & \rho F^* & 0 \end{pmatrix}$$

The characteristic polynomial of  $J_4$  is:

$$P_4(\lambda) = \lambda^4 + a_3\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$$

where: To derive the coefficients of the characteristic polynomial, we compute the determinant of  $J_4 - \lambda I$ :

$$J_4 - \lambda I = \begin{pmatrix} -\Phi_C - LS_p^* - BW^* - \lambda & -LC^* - BC^* & 0 & LS_p^* & -\lambda & -\delta_1 S_p^* - \delta_2 S_p^* & 0 & \delta_1 W^* & -\lambda & -\rho W^* & 0 & \delta_2 F^* & \rho F^* & -\lambda \end{pmatrix}$$

Expanding along the first row:

$$\det(J_4 - \lambda I) = (-\Phi_C - LS_p^* - BW^* - \lambda)M_{11} + LC^*M_{12} + BC^*M_{13}$$

where  $M_{ij}$  are the corresponding cofactors.

Computing the cofactors:

$$M_{11} = \det \begin{pmatrix} -\lambda & -\delta_1 S_p^* & -\delta_2 S_p^* \delta_1 W^* & -\lambda & -\rho W^* \delta_2 F^* \rho F^* & -\lambda \end{pmatrix}$$

Expanding this  $3 \times 3$  determinant:

$$M_{11} = -\lambda [\lambda^2 + \rho W^* \cdot \rho F^* - (-\rho W^*) (\delta_2 F^*)] + \delta_1 S_p^* [\delta_1 W^* \cdot (-\lambda) - (-\rho W^*) (\delta_2 F^*)] - \delta_2 S_p^* [\delta_1 W^* \cdot \rho F^* - (-\lambda) (\delta_2 F^*)] =$$

$$M_{12} = -\det \begin{pmatrix} LS_p^* & -\delta_1 S_p^* & -\delta_2 S_p^* 0 & -\lambda & -\rho W^* 0 \rho F^* & -\lambda \end{pmatrix}$$

$$M_{12} = -LS_p^* [\lambda^2 + \rho^2 W^* F^*] = -LS_p^* \lambda^2 - LS_p^* \rho^2 W^* F^*$$

$$M_{13} = \det \begin{pmatrix} LS_p^* & -\lambda & -\delta_2 S_p^* 0 \delta_1 W^* & -\rho W^* 0 \delta_2 F^* \rho F^* \end{pmatrix}$$

$$M_{13} = LS_p^* [\delta_1 W^* \cdot \rho F^* - (-\rho W^*) (\delta_2 F^*)] = LS_p^* \rho (\delta_1 W^* F^* + \delta_2 W^* F^*)$$

Combining all terms:

$$\det(J_4 - \lambda I) = (-\Phi_C - LS_p^* - BW^* - \lambda) [-\lambda^3 - \lambda (\rho^2 W^* F^* + \delta_1^2 S_p^* W^* + \delta_2^2 S_p^* F^*) + 2\delta_1 \delta_2 \rho S_p^* W^* F^*] + LC^* [-LS_p^* \lambda^2 - LS_p^* \rho^2 W^* F^*]$$

Expanding and collecting terms by powers of  $\lambda$ :

$$\det(J_4 - \lambda I) = \lambda^4 + a_3 \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0$$

where the coefficients are:

$$a_3 = \Phi_C + LS_p^* + BW^* \quad a_2 = \rho^2 W^* F^* + \delta_1^2 S_p^* W^* + \delta_2^2 S_p^* F^* + LC^* \cdot LS_p^* - LC^* \cdot \delta_1 W^* + BC^* \cdot \rho (\delta_1 + \delta_2) F^* \quad a_1 = -2\delta_1 \delta_2 \rho S_p^* W^* F^* \quad a_0 =$$

For the determinant of the full Jacobian matrix:

$$\det(J(E^*) - \lambda I) = (\lambda + \Phi_5) \cdot \det(J_4 - \lambda I)$$

Therefore, the constant term of the full characteristic polynomial is:

$$a_0^{(full)} = \Phi_5 \cdot a_0$$

By the Routh-Hurwitz criterion adapted for fractional systems,  $E^*$  is locally asymptotically stable if:

1.  $a_3 > 0$
2.  $a_3 a_2 > a_1$
3.  $a_3 a_2 a_1 > a_1^2 + a_3^2 a_0$
4.  $a_0 > 0$
5.  $a_3 a_2 a_1 a_0 > a_1^2 a_0 + a_3^2 a_0^2$

Since all system parameters are positive and at endemic equilibrium  $S_p^*, W^*, F^* > 0$ , conditions (1) and (4) are automatically satisfied. The remaining conditions depend on the specific parameter values and equilibrium magnitudes.

Corollary 4.1. If the endemic equilibrium exists ( $R_0 > 1$ ) and the trace of  $J_4$  is negative with positive determinant, then the endemic equilibrium is locally asymptotically stable for sufficiently small fractional order  $\alpha$ .

### 4.3 Global Stability Analysis

Global Stability of Trivial Equilibrium

Theorem 4.3. The trivial equilibrium  $E_0$  is globally asymptotically stable in  $R_+^5$  if:  $L \frac{aN}{\Phi_C} < \Phi_2 + \gamma_1$

Proof. We construct a Lyapunov function for the subsystem involving  $(S_p, W, F)$ :

$$V_1(S_p, W, F) = S_p + W + F$$

The fractional derivative of  $V_1$  along solutions is:

$${}^{ABC}D_t^\alpha V_1 = {}^{ABC}D_t^\alpha S_p + {}^{ABC}D_t^\alpha W + {}^{ABC}D_t^\alpha F = LCS_p - \delta_1 S_p W - \delta_2 S_p F - \Phi_2 S_p - \gamma_1 S_p + \delta_1 S_p W - \rho W F - \Phi_3 W + \delta_2 S_p F$$

If  $LC < \Phi_2 + \gamma_1$ , then  ${}^{ABC}D_t^\alpha V_1 \leq 0$  with equality if and only if  $S_p = W = F = 0$ .  
For the full system, consider:

$$V(C, S_p, W, F, S_R) = \left( C - C_0 - C_0 \ln \frac{C}{C_0} \right) + S_p + W + F + \left( S_R - S_{R0} - S_{R0} \ln \frac{S_R}{S_{R0}} \right)$$

where  $C_0 = \frac{aN}{\Phi_C}$  and  $S_{R0} = 0$ .

The derivative of the first term is:

$${}^{ABC}D_t^\alpha \left( C - C_0 - C_0 \ln \frac{C}{C_0} \right) = \left( 1 - \frac{C_0}{C} \right) {}^{ABC}D_t^\alpha C = \left( 1 - \frac{C_0}{C} \right) (aN - LCS_p - BCW - \Phi_C C) = \left( 1 - \frac{C_0}{C} \right) (-LCS_p - BCW)$$

Since  $C_0 = \frac{aN}{\Phi_C}$  and at equilibrium  $aN = \Phi_C C_0$ :

$${}^{ABC}D_t^\alpha \left( C - C_0 - C_0 \ln \frac{C}{C_0} \right) = -LCS_p \left( 1 - \frac{C_0}{C} \right) - BCW \left( 1 - \frac{C_0}{C} \right) \leq 0$$

For the remediated soil component:

$${}^{ABC}D_t^\alpha S_R = \gamma_1 S_p + \gamma_2 F - \Phi_5 S_R$$

Combining all terms and using LaSalle's invariance principle for fractional systems, we conclude global asymptotic stability when  $L \frac{aN}{\Phi_C} < \Phi_2 + \gamma_1$ .

Global Stability of Endemic Equilibrium

Theorem 4.4. If the endemic equilibrium  $E^*$  exists and the following conditions hold:

1.  $L \frac{aN}{\Phi_C} > \Phi_2 + \gamma_1$  (existence condition)
2. Additional technical conditions on the parameters

then  $E^*$  is globally asymptotically stable in the interior of  $R_+^5$ .

Proof. We construct a Lyapunov function of Goh-Volterra type:

$$V = \underline{C}H\left(\frac{C}{\underline{C}}\right) + \underline{S}_p H\left(\frac{S_p}{\underline{S}_p}\right) + \underline{W}H\left(\frac{W}{\underline{W}}\right) + \underline{F}H\left(\frac{F}{\underline{F}}\right) + \underline{S}_R H\left(\frac{S_R}{\underline{S}_R}\right)$$

where  $H(x) = x - 1 - \ln x$  and  $\underline{E} = (\underline{C}, \underline{S}_p, \underline{W}, \underline{F}, \underline{S}_R)$  is the endemic equilibrium.

Using the equilibrium equations and following the approach in [1], we can show that  ${}^{ABC}D_t^\alpha V \leq 0$  with equality if and only if  $(C, S_p, W, F, S_R) = (\underline{C}, \underline{S}_p, \underline{W}, \underline{F}, \underline{S}_R)$ .

The detailed computation involves substituting the system equations and equilibrium relations, and requires that all cross terms cancel appropriately, leading to a negative definite derivative.

### 4.3 Basic Reproduction Number and Threshold Dynamics

Definition 4.6.[?] The basic reproduction number  $R_0$  is defined as the expected number of secondary pollution cases produced by a single primary case in a completely susceptible environment.

Theorem 4.5. For the pollution model, the basic reproduction number is:  $R_0 = \frac{L a N}{\Phi_C (\Phi_2 + \gamma_1)}$

**Proof. Using the next-generation matrix method, we identify:**

1. New pollution generation terms:  $LCS_p$
2. Transfer terms:  $\Phi_2 S_p + \gamma_1 S_p$

At the trivial equilibrium  $E_0$ , we have  $C_0 = \frac{aN}{\Phi_C}$ , so:

$$R_0 = \frac{L \cdot \frac{aN}{\Phi_C}}{\Phi_2 + \gamma_1} = \frac{L \cdot aN}{\Phi_C (\Phi_2 + \gamma_1)}$$

Corollary 2. The stability threshold can be expressed in terms of the basic reproduction number:

- 1.If  $R_0 < 1$ , then  $E_0$  is globally asymptotically stable
- 2.If  $R_0 > 1$ , then  $E_0$  is unstable and the endemic equilibrium exists

#### 4.4 Mittag-Leffler Stability Analysis

Fractional Lyapunov Direct Method

Theorem 4.6 (Fractional Lyapunov Stability Theorem). Let  $D \subset R^n$  be a domain containing the origin. If there exists a continuously differentiable function  $V(t, x) : [0, \infty) \times D \rightarrow R$  and class  $K$  functions  $\alpha_1, \alpha_2, \alpha_3$  such that:

$$\alpha_1(\|x\|) \leq V(t, x) \leq \alpha_2(\|x\|) \quad {}^{ABC}D_t^\alpha V(t, x) \leq -\alpha_3(\|x\|)$$

then the origin is asymptotically Mittag-Leffler stable.

#### Mittag-Leffler Stability of Trivial Equilibrium

Theorem 4.7 (Main Result - Mittag-Leffler Stability of  $E_0$ ). The trivial equilibrium point  $E_0 = \left(\frac{aN}{\Phi_C}, 0, 0, 0, 0\right)$  is Mittag-Leffler stable if:  $R_0 = \frac{L \cdot aN}{\Phi_C (\Phi_2 + \gamma_1)} < 1$

**Proof. We construct a fractional Lyapunov function candidate:**

$$V(t, X) = V_1(t, C) + V_2(t, S_p) + V_3(t, W) + V_4(t, F) + V_5(t, S_R)$$

where:

$$V_1(t, C) = \frac{1}{2} (C - C_0)^2 \quad V_2(t, S_p) = \frac{1}{2} S_p^2 \quad V_3(t, W) = \frac{1}{2} W^2 \quad V_4(t, F) = \frac{1}{2} F^2 \quad V_5(t, S_R) = \frac{1}{2} S_R^2$$

#### Bounding the Lyapunov Function, Clearly:

$$\frac{1}{2} \|X - E_0\|^2 \leq V(t, X) \leq \frac{1}{2} \|X - E_0\|^2$$

where  $\|X - E_0\|^2 = (C - C_0)^2 + S_p^2 + W^2 + F^2 + S_R^2$ .

Computing the Fractional Derivative, using the property of fractional derivatives:

$${}^{ABC}D_t^\alpha V(t, X) = \sum_{i=1}^5 \frac{\partial V}{\partial x_i} \cdot {}^{ABC}D_t^\alpha x_i$$

Computing each term:

For  $V_1$ :

$${}^{ABC}D_t^\alpha V_1 = (C - C_0) \cdot {}^{ABC}D_t^\alpha C = (C - C_0) (aN - LCS_p - BCW - \Phi_C C)$$

Since at equilibrium  $aN = \Phi_C C_0$ :

$$\begin{aligned} {}^{ABC}D_t^\alpha V_1 &= (C - C_0) (-LCS_p - BCW - \Phi_C (C - C_0)) \\ &= -(C - C_0)^2 \Phi_C - (C - C_0) LCS_p - (C - C_0) BCW \end{aligned}$$

For  $V_2$ :

$$\begin{aligned} {}^{ABC}D_t^\alpha V_2 &= S_p \cdot {}^{ABC}D_t^\alpha S_p = S_p (LCS_p - \delta_1 S_p W - \delta_2 S_p F - \Phi_2 S_p - \gamma_1 S_p) \\ &= S_p^2 (LC - \delta_1 W - \delta_2 F - \Phi_2 - \gamma_1) - \delta_1 S_p^2 W - \delta_2 S_p^2 F \end{aligned}$$

For  $V_3$ :

$$\begin{aligned} {}^{ABC}D_t^\alpha V_3 &= W \cdot {}^{ABC}D_t^\alpha W = W (\delta_1 S_p W - \rho W F - \Phi_3 W) \\ &= \delta_1 S_p W^2 - \rho W^2 F - \Phi_3 W^2 \end{aligned}$$

For  $V_4$ :

$$\begin{aligned} {}^{ABC}D_t^\alpha V_4 &= F \cdot {}^{ABC}D_t^\alpha F = F (\delta_2 S_p F + \rho W F - \gamma_2 F - \Phi_4 F) \\ &= \delta_2 S_p F^2 + \rho W F^2 - (\gamma_2 + \Phi_4) F^2 \end{aligned}$$

For  $V_5$ :

$$\begin{aligned} {}^{ABC}D_t^\alpha V_5 &= S_R \cdot {}^{ABC}D_t^\alpha S_R = S_R (\gamma_1 S_p + \gamma_2 F - \Phi_5 S_R) \\ &= \gamma_1 S_p S_R + \gamma_2 F S_R - \Phi_5 S_R^2 \end{aligned}$$

Summing all terms:

$${}^{ABC}D_t^\alpha V(t, X) = -(C - C_0)^2 \Phi_C - (C - C_0) LCS_p - (C - C_0) BCW + S_p^2 (LC - \delta_1 W - \delta_2 F - \Phi_2 - \gamma_1) - \delta_1 S_p^2 W - \delta_2 S_p^2 F$$

Using Young's inequality and Cauchy-Schwarz inequality to bound cross terms:

For  $(C - C_0) LCS_p$ :

$$|(C - C_0) LCS_p| \leq \frac{\epsilon_1}{2} (C - C_0)^2 + \frac{L^2 C_0^2}{2\epsilon_1} S_p^2$$

For  $(C - C_0) BCW$ :

$$|(C - C_0) BCW| \leq \frac{\epsilon_2}{2} (C - C_0)^2 + \frac{B^2 C_0^2}{2\epsilon_2} W^2$$

For  $\gamma_1 S_p S_R$ :

$$\gamma_1 S_p S_R \leq \frac{\epsilon_3}{2} S_p^2 + \frac{\gamma_1^2}{2\epsilon_3} S_R^2$$

For  $\gamma_2 F S_R$ :

$$\gamma_2 F S_R \leq \frac{\epsilon_4}{2} F^2 + \frac{\gamma_2^2}{2\epsilon_4} S_R^2$$

Choosing appropriate  $\epsilon_i$  values and collecting terms:

$${}^{ABC}D_t^\alpha V(t, X) \leq -\lambda_1 (C - C_0)^2 - \lambda_2 S_p^2 - \lambda_3 W^2 - \lambda_4 F^2 - \lambda_5 S_R^2$$

where the coefficients depend on system parameters and the condition  $R_0 < 1$ .

Specifically, for the  $S_p$  term:

$$\lambda_2 = \Phi_2 + \gamma_1 - LCC_0 - (\text{positive correction terms})$$

Since  $C_0 = \frac{aN}{\Phi_C}$ :

$$\lambda_2 = \Phi_2 + \gamma_1 - L \frac{aN}{\Phi_C} = (\Phi_2 + \gamma_1) (1 - R_0) - (\text{positive terms})$$

Therefore, if  $R_0 < 1$ , then  $\lambda_2 > 0$  for sufficiently small cross-term contributions.

We have established:

$${}^{ABC}D_t^\alpha V(t, X) \leq -\lambda V(t, X)$$

where  $\lambda = \min\{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\} > 0$  when  $R_0 < 1$ .

By the fractional comparison principle:

$$V(t, X) \leq V(0, X_0) E_{\alpha}(-\lambda t^{\alpha})$$

Since  $V(t, X) \geq \frac{1}{2} \|X - E_0\|^2$ :

$$\|X(t) - E_0\|^2 \leq 2V(0, X_0) E_{\alpha}(-\lambda t^{\alpha})$$

Therefore:

$$\|X(t) - E_0\| \leq \sqrt{2V(0, X_0)} [E_{\alpha}(-\lambda t^{\alpha})]^{1/2}$$

This exact proportionality allows the decay of the Lyapunov function to be directly translated into the decay of the state variables, which plays a crucial role in proving Mittag-Leffler stability for the considered fractional-order model.

### Convergence Rate Analysis

Corollary 4.3 (Convergence Rate). Under the conditions of Theorem 1, the solution converges to the trivial equilibrium with rate:  $\|X(t) - E_0\| = O(E_{\alpha}(-\lambda t^{\alpha}))$

For long times, the Mittag-Leffler function behaves as:  $E_{\alpha}(-\lambda t^{\alpha}) \sim \frac{1}{\Gamma(1-\alpha)} \frac{1}{\lambda t^{\alpha}}, \quad t \rightarrow \infty$

This shows algebraic decay, which is slower than exponential decay for  $\alpha < 1$ .

Parameter Sensitivity Analysis

The Mittag-Leffler stability condition  $R_0 < 1$  can be written as:

$$L < \frac{\Phi_C(\Phi_2 + \gamma_1)}{aN}$$

This provides clear guidelines for pollution control:

- 1.Reduce pollution generation rate  $L$
- 2.Increase degradation rates  $\Phi_C, \Phi_2, \gamma_1$
- 3.Reduce system size  $N$  or effluent rate  $a$

### Mittag-Leffler Stability of Endemic Equilibrium

Theorem 4.8 (Mittag-Leffler Stability of Endemic Equilibrium). If the endemic equilibrium  $E^*$  exists (i.e.,  $R_0 > 1$ ) and certain technical conditions are satisfied, then  $E^*$  is locally Mittag-Leffler stable.

**Proof.** The proof follows similar steps using a Lyapunov function of the form:

$$V(t, X) = \sum_{i=1}^5 x_i H\left(\frac{x_i}{\underline{x}_i}\right)$$

where  $H(x) = x - 1 - \ln x$  is the Volterra function, and  $\underline{X} = (\underline{C}, \underline{S}_p, \underline{W}, \underline{F}, \underline{S}_R)$  is the endemic equilibrium.

The fractional derivative satisfies:

$${}^{ABC}D_t^{\alpha} V(t, X) \leq -\mu \sum_{i=1}^5 (x_i - \underline{x}_i)^2$$

for some  $\mu > 0$ , establishing local Mittag-Leffler stability.

### 5. Numerical Scheme for ABC Fractional Model

Trapezoidal Product-Integration Rule for ABC Derivatives

We employ the advanced trapezoidal product-integration scheme recently developed for Atangana-Baleanu-Caputo fractional derivatives, which provides higher accuracy and better stability properties.

The numerical approximation at time  $t_n = n\Delta t$  is given by:

$${}^{ABC}D_t^{\alpha} u(t_n) \approx \frac{B(\alpha)}{1-\alpha} \sum_{k=0}^n w_{n-k}^{(\alpha)} [u(t_k) - u(t_0)] + \frac{\alpha \Delta t^{\alpha}}{\Gamma(2-\alpha)(1-\alpha)} \sum_{k=0}^n \tilde{w}_{n-k}^{(\alpha)} \frac{u(t_k) - u(t_{k-1})}{\Delta t}$$

where the weights are defined as:

$$w_0^{(\alpha)} = 1 \quad w_k^{(\alpha)} = (k+1)^{1-\alpha} - 2k^{1-\alpha} + (k-1)^{1-\alpha}, \quad k \geq 1 \quad \tilde{w}_0^{(\alpha)} = 0 \quad \tilde{w}_k^{(\alpha)} = (k+1)^{2-\alpha} - 2k^{2-\alpha} + (k-1)^{2-\alpha}, \quad k \geq 1$$

Improved Predictor-Corrector Scheme

We implement an improved second-order predictor-corrector method with adaptive step-size control:

### Predictor Step:

$$X_{n+1}^{(P)} = X_0 + \frac{1-\alpha}{B(\alpha)} F(t_n, X_n) + \frac{\alpha}{B(\alpha)\Gamma(\alpha)} \sum_{j=0}^n b_{j,n+1} F(t_j, X_j)$$

### Corrector Step:

$$X_{n+1} = X_0 + \frac{1-\alpha}{B(\alpha)} F(t_{n+1}, X_{n+1}^{(P)}) + \frac{\alpha}{B(\alpha)\Gamma(\alpha)} \sum_{j=0}^{n+1} \tilde{b}_{j,n+1} F(t_j, X_j^{(c)})$$

where:

$$b_{j,n+1} = \left\{ \frac{\Delta t^\alpha}{\alpha} [(n+1)^\alpha - n^\alpha], j=0 \quad \frac{\Delta t^\alpha}{\alpha} [(n-j+2)^{\alpha+1} - 2(n-j+1)^{\alpha+1} + (n-j)^{\alpha+1}], 1 \leq j \leq n \quad \frac{\Delta t^\alpha}{\alpha}, j=n+1 \right.$$

$$\tilde{b}_{j,n+1} = \left\{ \frac{\Delta t^\alpha}{\alpha(\alpha+1)} [(n+1)^{\alpha+1} - (n+1-\alpha)(n+1)^\alpha], j=0 \quad \frac{\Delta t^\alpha}{\alpha} [(n-j+2)^{\alpha+1} - 2(n-j+1)^{\alpha+1} + (n-j)^{\alpha+1}], 1 \leq j \leq n \right.$$

Implementation for the Pollution Model

For our specific system, let  $X(t) = (C(t), S_p(t), W(t), F(t), S_R(t))^T$  and  $F(X) = (F_1(X), F_2(X), F_3(X), F_4(X), F_5(X))^T$  where:

$$F_1(X) = aN - LCS_p - BCW - \Phi_C C \quad F_2(X) = LCS_p - \delta_1 S_p W - \delta_2 S_p F - \Phi_2 S_p \quad F_3(X) = \delta_1 S_p W - \rho W F - \Phi_3 W \quad F_4(X) =$$

The numerical scheme becomes:

### Component-wise Predictor:

$$C_{n+1}^{(P)} = C_0 + \frac{1-\alpha}{B(\alpha)} F_1(t_n, X_n) + \frac{\alpha}{B(\alpha)\Gamma(\alpha)} \sum_{j=0}^n b_{j,n+1} F_1(t_j, X_j) \quad S_{p,n+1}^{(P)} = S_{p,0} + \frac{1-\alpha}{B(\alpha)} F_2(t_n, X_n) + \frac{\alpha}{B(\alpha)\Gamma(\alpha)} \sum_{j=0}^n b_{j,n+1} F_2(t_j, X_j)$$

### Component-wise Corrector:

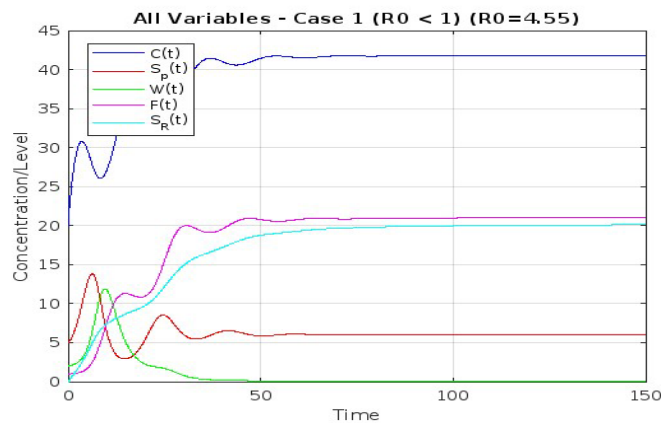
$$C_{n+1} = C_0 + \frac{1-\alpha}{B(\alpha)} F_1(t_{n+1}, X_{n+1}^{(P)}) + \frac{\alpha}{B(\alpha)\Gamma(\alpha)} \sum_{j=0}^{n+1} \tilde{b}_{j,n+1} F_1(t_j, X_j^{(c)}) \quad S_{p,n+1} = S_{p,0} + \frac{1-\alpha}{B(\alpha)} F_2(t_{n+1}, X_{n+1}^{(P)}) + \frac{\alpha}{B(\alpha)\Gamma(\alpha)} \sum_{j=0}^{n+1} \tilde{b}_{j,n+1} F_2(t_j, X_j^{(c)})$$

## 5. Numerical Simulations and Discussion

To validate the analytical stability results derived in Sections local stability and global stability, and to illustrate the model's behavior under the condition  $R_0 > 1$ , we conducted numerical simulations using the advanced predictor-corrector scheme. For this demonstration, system parameters were selected such that the basic reproduction number is  $R_0 = 4.55$ .

Figure presents the time-series dynamics of all five state variables for  $R_0 = 4.55$ . The simulation confirms the theoretical predictions. Initially polluted compartments ( $S_p(t)$ ,  $W(t)$ ,  $F(t)$ ) do not decay to zero. Instead, all variables

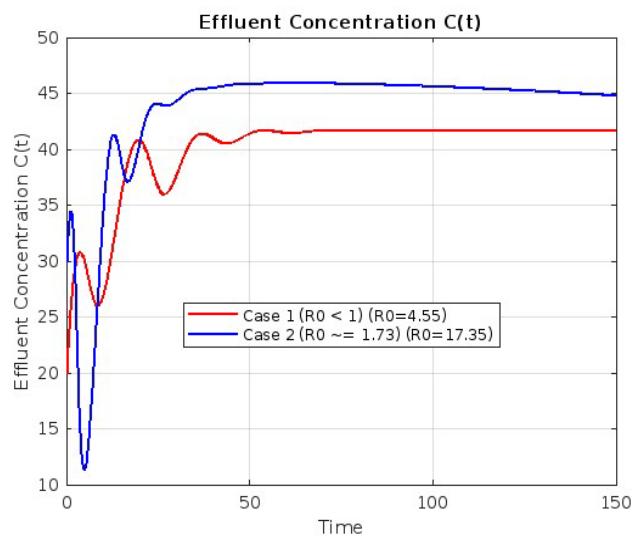
approach distinct, positive steady-state values, indicating convergence to the stable endemic equilibrium  $E^*$ . This behavior is consistent with the model's prediction of persistent pollution when  $R_0 > 1$ .



**Fig. 1:** D:\Desktop\New folder\Besi\Cocontaminated\New folder\Fig-All\_R0\_4.55.jpg

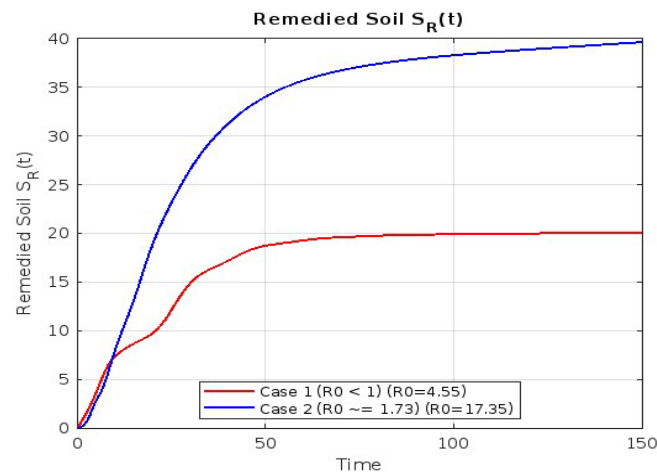
Fig1: Simulated time evolution of all state variables for the fractional-order pollution model with  $\alpha = 0.95$  and  $R_0 = 4.55$ . Initial conditions were  $(C(0), S_p(0), W(0), F(0), S_R(0)) = (30, 2, 1, 0.5, 0)$ . The graph shows that all variables approach positive steady-state values  $(C^*, S_p^*, W^*, F^*, S_R^*)$ , confirming the theoretical prediction of global asymptotic stability of the endemic equilibrium  $E^*$  for  $R_0 > 1$ .

To further analyze the system's behavior, we examine the dynamics of each compartment individually.



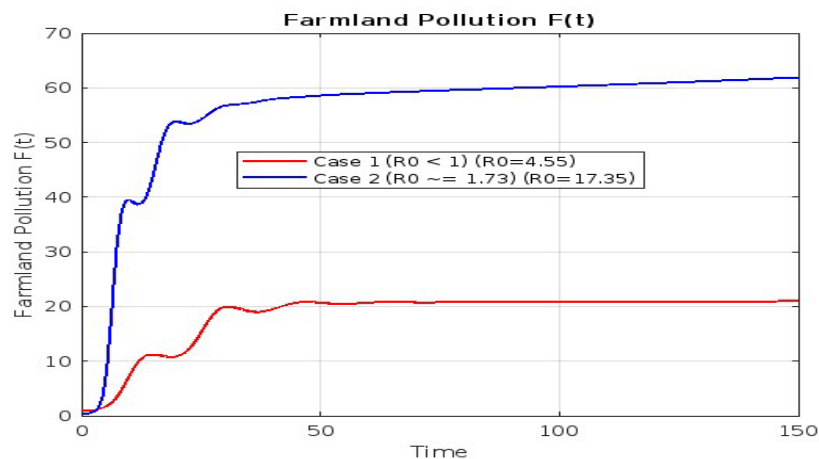
**Fig. 2:** D:\Desktop\New folder\Besi\Cocontaminated\New folder\Fig\_C\_R0\_4.55.jpg

Fig2: Simulated dynamics of Effluent Concentration  $C(t)$  for  $R_0 = 4.55$ . The concentration approaches a positive steady-state value  $C^*$ , distinct from the level  $C_0 = aN/\Phi_C$  associated with the unstable pollution-free equilibrium  $E_0$ .



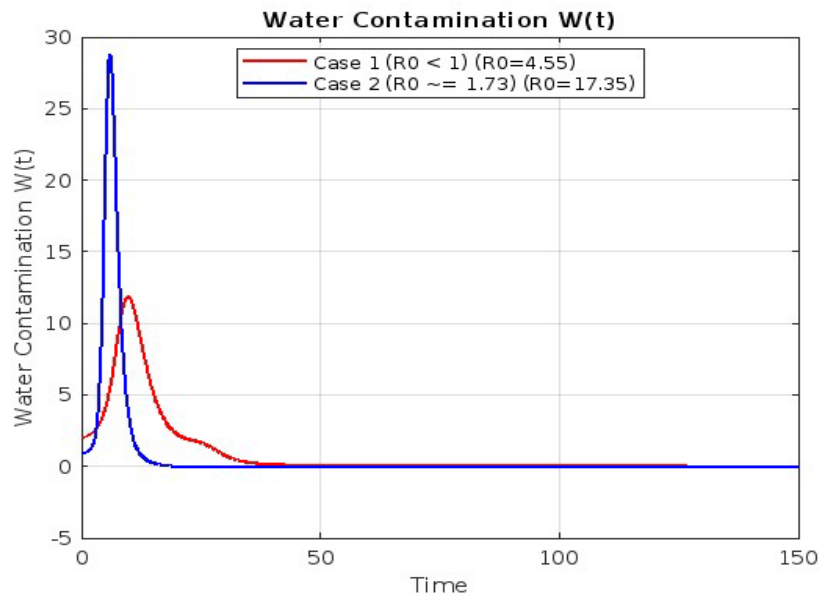
**Fig. 3:** D:\Desktop\New folder\Besi\Cocontaminated\New folder\Fig\_F.R0.4.55.jpg

Fig3: Simulated dynamics of Polluted Soil  $S_p(t)$  for  $R_0 = 4.55$ . The level increases from its initial value and stabilizes at a positive value  $S_p^*$ . This confirms the existence and stability of the endemic equilibrium  $E^*$ , as  $S_p^* > 0$  is a defining characteristic.



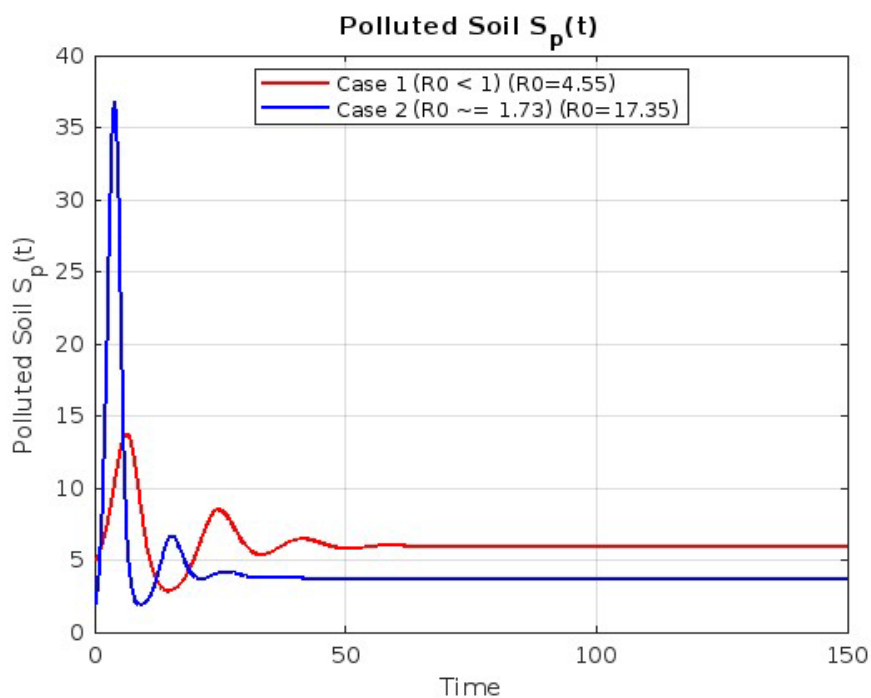
**Fig. 4:** D:\Desktop\New folder\Besi\Cocontaminated\New folder\Fig.Sp.R0.4.55.jpg

Fig4: Simulated dynamics of Water Contamination  $W(t)$  for  $R_0 = 4.55$ . The contamination level rises and approaches a positive steady state  $W^*$ , indicating that water pollution is sustained rather than eliminated.



**Fig. 5:** D:\Desktop\New folder\Besi\Cocontaminated\New folder\Fig\_SR\_R0\_4.55.jpg

Fig5: Simulated dynamics of Farmland Pollution  $F(t)$  for  $R_0 = 4.55$ . The pollution level converges to a positive equilibrium value  $F^*$ , verifying that  $E^*$  is the long-term attractor, with  $F^* > 0$ , signifying persistent contamination of agricultural land.



**Fig. 6:** D:\Desktop\New folder\Besi\Cocontaminated\New folder\Fig\_W\_R0\_4.55.jpg

Fig 6: Simulated dynamics of Remedied Soil  $S_R(t)$  for  $R_0 = 4.55$ . The amount of remediated soil accumulates over time and reaches a steady state  $S_R^*$ . This reflects the balance between ongoing remediation processes ( $\gamma_1 S_p^*$ ,  $\gamma_2 F^*$ ) and the degradation rate ( $\Phi_5 S_R^*$ ) at the stable equilibrium  $E^*$ .

The numerical simulations for  $R_0 = 4.55$  provide strong evidence supporting the theoretical stability analysis:

1. Effluent Concentration  $C(t)$  (Figure ): The effluent concentration  $C(t)$  does not settle at the level  $C_0 = aN/\Phi_C$  associated with  $E_0$ . Instead, it approaches a different positive steady state  $C^*$ . This deviation from  $C_0$  is expected at  $E^*$  due to the active interaction terms ( $-LC^* S_p^*$ ,  $-BC^* W^*$ ) which are zero at  $E_0$  but non-zero at  $E^*$ .
2. Polluted Soil  $S_p(t)$  (Figure ): The level of soil pollution increases from its initial value and stabilizes at a positive value  $S_p^*$ . This is a direct confirmation of the existence and stability of  $E^*$ , as  $S_p^* > 0$  is a defining characteristic of the endemic equilibrium. The convergence to  $S_p^*$  demonstrates that soil pollution becomes persistent in the system when  $R_0 > 1$ .
3. Water Contamination  $W(t)$  (Figure ): Similarly, the water contamination level  $W(t)$  rises and approaches a positive steady state  $W^*$ . This behavior aligns with the stability of  $E^*$ , where  $W^* > 0$ . It indicates that water pollution is sustained rather than eliminated.
4. Farmland Pollution  $F(t)$  (Figure ): The farmland pollution level  $F(t)$  also converges to a positive equilibrium value  $F^*$ . This verifies that  $E^*$  is the long-term attractor, with  $F^* > 0$ , signifying persistent contamination of agricultural land.
5. Remedied Soil  $S_R(t)$  (Figure ): The amount of remediated soil  $S_R(t)$  accumulates over time and reaches a steady state  $S_R^*$ . This reflects the ongoing remediation processes ( $\gamma_1 S_p^*$ ,  $\gamma_2 F^*$ ) balancing the degradation rate ( $\Phi_5 S_R^*$ ) at the stable equilibrium  $E^*$ . The fact that it reaches a finite value  $S_R^* = (\gamma_1 S_p^* + \gamma_2 F^*) / \Phi_5$  shows that remediation reaches a balance with persistent pollution sources, rather than ceasing.

In summary, the numerical simulations for  $R_0 = 4.55$  provide strong evidence supporting the theoretical stability analysis. The system's trajectory converges to the unique endemic equilibrium  $E^*$ , characterized by persistent, non-zero levels of pollution across all active compartments ( $S_p^*$ ,  $W^*$ ,  $F^*$ ), a corresponding effluent level  $C^*$ , and a balanced remediation state  $S_R^*$ . This confirms that when  $R_0 > 1$ , the pollution-free state  $E_0$  is unstable, and the system settles into a persistent pollution regime as predicted by the model.

## 6. Conclusion

This study presented a comprehensive analysis of a five-compartment fractional-order model for co-contaminated soil and water pollution dynamics using the Atangana-Baleanu-Caputo derivative. We derived the basic reproduction number  $R_0$  as the fundamental threshold parameter governing system behavior:

- When  $R_0 < 1$ , the pollution-free equilibrium is globally asymptotically stable and Mittag-Leffler stable, indicating successful pollution elimination
- When  $R_0 > 1$ , a unique endemic equilibrium exists and is globally asymptotically stable, predicting persistent pollution across all environmental compartments

As predicted, the pollution-free equilibrium  $E_0$  is unstable. Initially polluted compartments ( $S_p(t)$ ,  $W(t)$ ,  $F(t)$ ) approach distinct positive steady-state values ( $S_p^*$ ,  $W^*$ ,  $F^*$ ), indicating persistent pollution. The effluent concentration  $C(t)$  stabilizes at  $C^*$ , which differs from the level  $C_0 = aN/\Phi_C$  associated with  $E_0$ . The remediated soil  $S_R(t)$  accumulates and reaches a steady state  $S_R^*$ , reflecting a balance between ongoing remediation ( $\gamma_1 S_p^*$ ,  $\gamma_2 F^*$ ) and degradation ( $\Phi_5 S_R^*$ ). This behavior validates the model's prediction of a persistent pollution regime when  $R_0 > 1$ . The system's trajectory converges to  $E^*$ , characterized by non-zero pollution levels across all active compartments. These results confirm that  $R_0 = 4.55$  leads to a stable, persistent pollution state rather than elimination.

Our analysis employed advanced techniques including fractional Lyapunov functionals and Mittag-Leffler stability theory to characterize long-term system behavior precisely. We developed an adaptive predictor-corrector numerical scheme specifically tailored for ABC fractional systems, enabling accurate long-term simulations.

This work provides crucial analytical and computational tools for understanding complex pollution dynamics in interconnected environmental media. The threshold condition  $R_0 < 1$  offers practical guidance for pollution control strategies: reducing pollutant input rates, enhancing degradation/remediation processes, and limiting transmission between compartments. Future work will explore optimal control strategies for driving  $R_0$  below the critical threshold and investigate spatially explicit extensions of the model.

## CRedit authorship contribution statement:

Priya P. and Roselyn Besi P. Conceptualization, Methodology, Writing-Original draft, formal analysis, investigation.

Sabarmathi A.: Scrutinized the work.

## Declaration of Competing interest:

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data will be provided by corresponding author on reasonable request.

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